

Total No. of Questions : 12]

SEAT No. :

P-9347

[Total No. of Pages : 3

[6181]-582

B.E. (Mechanical Engineering)

INDUSTRIAL ENGINEERING

(2015 Pattern) (Semester - II) (402049B) (Elective - III)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answers should be written in one answer book.
- 2) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10, Q.11 or Q.12.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Figures to the right side indicate full marks.
- 5) Assume suitable data, if necessary.

Q1) Define Industrial engineering. Explain in brief various function of Industrial engineering? **[8]**

OR

Q2) Discuss in details factors that affect productivity. **[8]**

Q3) Explain with proper example Multiple activity chart. **[8]**

OR

Q4) a) Differentiate between Value Engineering and Value Analysis. **[4]**

b) Show and describe method study symbols for recording the facts. **[4]**

Q5) A work sampling study was conducted to establish the standard time for an operation. The observations of the study given below; **[6]**

Total no. of operations = 160

Manual (hand controlled work = 14

Machine controlled work = 106

Average performance rating = 80%

No. of parts produced = 36

Allowance for personal needs and fatigue = 10%

Study conducted for 3 days

Available working hours/day = 8hrs

Calculate standard time per piece.

OR

P.T.O.

- Q6)** The elemental timings are given below along with the respective ratings. Assuming rest and personal allowances as 12% and contingency allowance as 2%, calculate the standard time for the operation. **[6]**

Element	Observed Time	Rating	Remark
A	0.2	90	
B	0.05	80	
C	0.03	100	
D	0.78	100	
E	0.06	100	
F	0.05	100	
G	0.02	85	Once in 5 Pieces
H	0.06	80	
I	0.10	90	
J	0.04	90	Once in 20 Pieces

- Q7)** a) Explain in details MRP and MRP - II. **[8]**
 b) ABC Company uses exponential smoothing with $\alpha = 0.15$ for estimating a demand. Following table shows the demand for 6 months. The forecast for June is 500 units. Estimate the forecast for December. **[8]**

Month	Actual Demand	Forecast
June	450	500
July	505	
August	516	
September	488	
October	467	
November	554	

OR

- Q8)** a) Short Note on ERP with Block Diagram. **[8]**
 b) Explain in brief various functions of Production Planning and Control. **[8]**

- Q9)** a) Write short note on **[8]**
 i) Material Flow Pattern
 ii) Line Balancing
 b) Define plant layout. What are various factors involved in selection in selection of site for good plant. **[8]**

OR

Q10) a) Enlist material handling equipments. Describe any three material handling equipment with suitable illustration. [8]

b) A manufacturing company requires 9500 units per year. Ordering cost is Rs. 125 per order & carrying cost is 20%. Purchase price per unit is Rs. 45. Determine : [8]

- i) EOQ
- ii) Optimum no. of orders
- iii) Total cost including acquisition of material

Q11) a) A Company has given following information [8]

Data	Rs.
Sales	1,50,000
Variable Overheads	1,20,000
Gross Profit	60,000
Fixed Overheads	20,000
Net Profit	40,000

Find

- i) P/V ratio
 - ii) BEP
 - iii) Net Profit when the sales are Rs. 4,00,000
 - iv) Sales required to earn a profit of Rs 80,000
- b) What is importance of standard costing? Explain different elements of costs. [8]

OR

Q12) a) Explain break-even analysis with neat chart? What are assumptions made in break-even analysis? What are different applications of break-even analysis. [8]

b) Explain in brief different functions of human resource department. Comment on concept of key result areas and performance appraisal of employee. [8]

